

Homological Algebra Exercises

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1 Problem sheet 1

Problem 1.1.

(i) To show

$$\begin{aligned}\epsilon : R[x] &\rightarrow R \\ \alpha &\mapsto \alpha(0)\end{aligned}$$

is a ring homomorphism. Let $\alpha, \beta \in R[x]$ we have

$$\epsilon(\alpha + \beta) = (\alpha + \beta)(0) = \alpha(0) + \beta(0) = \epsilon(\alpha) + \epsilon(\beta).$$

Similarly,

$$\epsilon(\alpha\beta) = (\alpha\beta)(0) = \sum_{r=0}^0 \alpha(r)\beta(-r) = \alpha(0)\beta(0) = \epsilon(\alpha)\epsilon(\beta).$$

In addition,

$$\epsilon(1) = 1(0) = 1$$

so ϵ is a ring homomorphism.

(ii) To show the sequence

$$0 \longrightarrow R[x] \xrightarrow{\times x} R[x] \xrightarrow{\epsilon} R \longrightarrow 0$$

is exact. We first show that multiplication by x is injective and ϵ is surjective. Define

$$\begin{aligned}f : R[x] &\rightarrow R[x] \\ \alpha(x) &\mapsto \alpha(x)x\end{aligned}$$

It's trivial that $\ker f = \{0\}$ and it should be clear that ϵ is surjective by definition. Now, to show that $\text{Im}(f) = \ker \epsilon$. Notice

$$\ker \epsilon = \{\alpha(x) \in R[x] \mid \alpha(0) = 0\}$$

i.e. if $\alpha(x) \in \ker \epsilon$ then

$$\alpha(x) = \sum_{r=1}^{\deg(\alpha)} \alpha(r)x^r = x \sum_{r=1}^{\deg(\alpha)} \alpha(r)x^{r-1} = x \underbrace{\sum_{r=0}^{\deg(\alpha)-1} \alpha(r+1)x^r}_{\in R[x]}$$

Hence, $\ker \epsilon \subseteq \text{Im}(f)$ and the other inclusion should be clear. Hence, the sequence is exact. Alternatively, a polynomial has zero constant term if and only if it is divisible by x .

- (iii) ($\implies$) : Suppose $R[x]$ be an integral domain, since R is a subring of $R[x]$ with unit so it's clear that R is an integral domain.
- ($\impliedby$) : Suppose R be an integral domain. We shall proceed by induction to show that $R[x]$ is an integral domain. Let $\alpha(x), \beta(x) \in R[x]$ be two constant polynomials then it's clear that $\alpha(x)\beta(x) = 0$ implies $\alpha(x) = 0$ or $\beta(x) = 0$ since R is an integral domain. Let $n \geq 1$, suppose for all $\alpha(x), \beta(x) \in R[x]$ where $\max\{\deg(\alpha), \deg(\beta)\} \leq n$, $\alpha(x)\beta(x) = 0$ implies either $\alpha(x) = 0$ or $\beta(x) = 0$. Let $\alpha_1(x), \beta_1(x) \in R[x]$ such that $\max\{\deg(\alpha_1), \deg(\beta_1)\} = n + 1$ and $\alpha_1(x)\beta_1(x) = 0$. In particular,

$$\epsilon(\alpha_1(x)\beta_1(x)) = \epsilon(\alpha_1(x))\epsilon(\beta_1(x)) = 0.$$

Since R is an integral domain, it follows that $\epsilon(\alpha_1(x)) = 0$ or $\epsilon(\beta_1(x)) = 0$. WLOG, suppose $\epsilon(\alpha_1(x)) = 0$. By exactness, we have seen previously that $\ker \epsilon = \text{Im}(f)$, since $\alpha_1(x) \in \ker \epsilon$, there exists some $\alpha(x) \in R[x]$ such that $\alpha_1(x) = x\alpha(x)$ so that $x\alpha(x)\beta_1(x) = 0$. But multiplication by x is injective so $\alpha(x)\beta_1(x) = 0$ where $\deg(\alpha) \leq n$. If $\alpha(x) = 0$ we are done. Now, we assume that $\alpha(x) \neq 0$. Again, applying

$$\epsilon(\alpha(x)\beta_1(x)) = \epsilon(\alpha(x))\epsilon(\beta_1(x)) = 0$$

Then either $\epsilon(\alpha(x)) = 0$ or $\epsilon(\beta_1(x)) = 0$. If $\epsilon(\beta_1(x)) = 0$ then by the same argument as before, there exists some $\beta(x) \in R[x]$ such that $\beta_1(x) = x\beta(x)$ and $\deg(\beta) \leq n$. Injectivity implies $\alpha(x)\beta(x) = 0$ and we are done by the inductive hypothesis. Hence, we assume that $\epsilon(\beta(x)) \neq 0$. Then there exists some $\alpha_2(x) \in R[x]$ such that $\alpha(x) = x\alpha_2(x)$. Injectivity implies $\alpha_2(x)\beta_1(x) = 0$. We could repeat this process to construct a sequence of polynomial $\{\alpha_i(x)\} \subset R[x]$ such that $\alpha_i(x) = x\alpha_{i-1}(x)$. But since n is finite, there exists some $N \in \mathbb{N}$ such that $\alpha_N = C$ where C is a constant. Then eventually we have $\beta_1(x) = 0$. Thus, it follows that the statement is true for all $n \in \mathbb{N}$ i.e. $R[x]$ is an integral domain.

Problem 1.2.

- (i) The Criteria for $\text{Hom}_R(M, N)$ to be an R -module when R is commutative is easy to check. We shall explain why it's no longer true when R is not commutative. For $r \in R$, $f \in \text{Hom}_R(M, N)$, we claim that $fr \notin \text{Hom}_R(M, N)$. For $m \in M$, and pick some $s \in R$ such that $rs \neq sr$, if $fr \in \text{Hom}_R(M, N)$, linearity implies

$$(fr)(ms) = (f(ms))r = (f(m)s)r = f(m)(sr)$$

$$(fr)(ms) = ((fr)(m))s = (f(m)r)s = f(m)(rs)$$

i.e. $sr = rs$, a contradiction.

- (ii) Consider the map

$$\begin{aligned} \varphi : \text{Hom}_R(R, M) &\rightarrow M \\ f &\mapsto f(1) \end{aligned}$$

where $f \in \text{Hom}_R(R, M)$ and 1 is the unit in R . It is enough to show that φ is an isomorphism. We first check it's a homomorphism: let $f, g \in \text{Hom}_R(R, M)$ then

$$\varphi(f + g) = (f + g)(1) = f(1) + g(1) = \varphi(f) + \varphi(g).$$

And for any $r \in R$ we have

$$\varphi(fr) = (fr)(1) = f(1)r = \varphi(f)r$$

so φ is a homomorphism. To see it's injective: let $f, g \in \text{Hom}_R(R, M)$ where $f(1) = g(1)$. For any $r \in R$:

$$f(r) = f(1)r = g(1)r = g(r)$$

so $f = g$ on R . To show it's surjective, let $m \in M$ and define

$$\begin{aligned} f_m : R &\rightarrow M \\ r &\mapsto mr \end{aligned}$$

We claim that this is an R -module homomorphism. Indeed, for $r_1, r_2 \in R$:

$$f_m(r_1 + r_2) = m(r_1 + r_2) = mr_1 + mr_2 = f_m(r_1) + f_m(r_2).$$

And for any $r \in R$:

$$f_m(r_1r) = m(r_1r) = (mr_1)r = f_m(r_1)r.$$

So $f_m \in \text{Hom}_R(R, M)$ and notice $\varphi(f_m) = f_m(1) = m$ which completes the proof of surjectivity.

(iii) Consider the map

$$\begin{aligned} \varphi : \text{Hom}_R(M_1, N) \oplus \text{Hom}_R(M_2, N) &\rightarrow \text{Hom}_R(M_1 \oplus M_2, N) \\ (f, g)(x, y) &\mapsto f(x) + g(y) \end{aligned}$$

where $f \in \text{Hom}_R(M_1, N)$, $g \in \text{Hom}_R(M_2, N)$ and $(x, y) \in M_1 \oplus M_2$. Notice this map is well-defined: given $(x_1, y_1), (x_2, y_2) \in M_1 \oplus M_2$ and $r \in R$ we have

$$\begin{aligned} \varphi(f, g)((x_1, y_1) + (x_2, y_2)r) &= \varphi(f, g)(x_1 + x_2r, y_1 + y_2r) \\ &= f(x_1 + x_2r) + g(y_1 + y_2r) = f(x_1) + f(x_2)r + g(y_1) + g(y_2)r \\ &= (f(x_1) + g(y_1)) + (f(x_2) + g(y_2))r = \varphi(f, g)(x_1, y_1) + \varphi(f, g)(x_2, y_2)r \end{aligned}$$

Therefore, $\varphi(f, g) \in \text{Hom}_R(M_1 \oplus M_2, N)$. We claim that this map is an isomorphism.

(Homomorphism) Let $(f_1, g_1), (f_2, g_2) \in \text{Hom}_R(M_1, N) \oplus \text{Hom}_R(M_2, N)$ and $r \in R$, for any $(x, y) \in M_1 \oplus M_2$:

$$\begin{aligned} \varphi((f_1, g_1) + (f_2, g_2)r)(x, y) &= \varphi((f_1 + f_2r, g_1 + g_2r))(x, y) \\ &= (f_1 + f_2r)(x) + (g_1 + g_2r)(y) = f_1(x) + f_2(x)r + g_1(y) + g_2(y)r \\ &= (f_1(x) + g_1(y)) + (f_2(x) + g_2(y))r = \varphi(f_1, g_1)(x, y) + \varphi(f_2, g_2)(x, y)r \end{aligned}$$

(Isomorphism) Suppose $\varphi(f_1, g_1)(x, y) = \varphi(f_2, g_2)(x, y)$ for all $(x, y) \in M_1 \oplus M_2$. Consider the points $(x, 0_{M_2})$ with $x \in M_1$ we have

$$f_1(x) = \varphi(f_1, g_1)(x, 0_{M_2}) = \varphi(f_2, g_2)(x, 0_{M_2}) = f_2(x)$$

for all $x \in M_1$ i.e. $f_1 = f_2$. Similar argument shows $g_1 = g_2$ so φ is injective. For surjectivity, let $\eta \in \text{Hom}_R(M_1 \oplus M_2, N)$ and for $f : M_1 \rightarrow N$ and $g : M_2 \rightarrow N$ define

$$f(x) = \eta(x, 0_{M_2}) \text{ and } g(y) = \eta(0_{M_1}, y).$$

It is enough to show that $f \in \text{Hom}_R(M_1, N)$ and $g \in \text{Hom}_R(M_2, N)$. Indeed, for $x_1, x_2 \in M_1$ and $r \in R$:

$$\begin{aligned} f(x_1 + x_2)r &= \eta(x_1 + x_2r, 0_{M_2}) = \eta((x_1, 0_{M_2}) + (x_2, 0_{M_2})r) \\ &= \eta(x_1, 0_{M_2}) + \eta(x_2, 0_{M_2})r \\ &= f(x_1) + f(x_2)r \end{aligned}$$

Similar argument shows $g \in \text{Hom}_R(M_2, N)$. Now, notice that for any $(x, y) \in M_1 \oplus M_2$:

$$\varphi(f, g)(x, y) = f(x) + g(y) = \eta(x, 0_{M_2}) + \eta(0_{M_1}, y) = \eta(x, y)$$

which shows surjectivity.

(iv) In this question, we need to use the fact that the following diagram

$$\begin{array}{ccc} & N_1 \oplus N_2 & \\ i_1 \nearrow & & \nwarrow i_2 \\ N_1 & \xleftarrow{\pi_1} & N_2 \end{array}$$

commutes. In particular,

$$\pi_1 \circ i_1 = \text{id}_{N_1}, \quad \pi_2 \circ i_2 = \text{id}_{N_2}, \quad \pi_1 \circ i_2 = \pi_2 \circ i_1 = 0$$

and $i_1 \circ \pi_1 + i_2 \circ \pi_2 = \text{id}_{N_1 \oplus N_2}$. Define

$$\begin{aligned} \varphi : \text{Hom}_R(M, N_1 \oplus N_2) &\rightarrow \text{Hom}_R(M, N_1) \oplus \text{Hom}_R(M, N_2) \\ f &\mapsto (\pi_1 \circ f, \pi_2 \circ f) \end{aligned}$$

and

$$\begin{aligned} \psi : \text{Hom}_R(M, N_1) \oplus \text{Hom}_R(M, N_2) &\rightarrow \text{Hom}_R(M, N_1 \oplus N_2) \\ (g, h) &\mapsto i_1 \circ g + i_2 \circ h \end{aligned}$$

φ is well-defined since for $x, y \in M$, $r \in R$:

$$\begin{aligned} \varphi(f(x + yr)) &= (\pi_1 \circ f(x + yr), \pi_2 \circ f(x + yr)) \\ &= (\pi_1(f(x) + f(y)r), \pi_2(f(x) + f(y)r)) \\ &= (\pi_1(f(x)) + \pi_1(f(y)r), \pi_2(f(x)) + \pi_2(f(y)r)) \\ &= (\pi_1(f(x)), \pi_2(f(x))) + (\pi_1(f(y)), \pi_2(f(y)))r \\ &= \varphi(f(x)) + \varphi(f(y))r \end{aligned}$$

Similarly, ψ is well-defined: for $x, y \in M, r \in R$:

$$\begin{aligned}\psi(g, h)(x + yr) &= i_1(g(x + yr)) + i_2(h(x + yr)) \\ &= i_1(g(x) + g(y)r) + i_2(h(x) + h(y)r) \\ &= i_1(g(x)) + i_1(g(y))r + i_2(h(x)) + i_2(h(y))r \\ &= \psi(g, h)(x) + \psi(g, h)(y)r\end{aligned}$$

To show φ is a homomorphism, let $f, g \in \text{Hom}_R(M, N_1 \oplus N_2)$, $x \in M$ and $r \in R$:

$$\begin{aligned}\varphi(f + gr)(x) &= (\pi_1((f + gr)(x)), \pi_2((f + gr)(x))) \\ &= (\pi_1(f(x)) + \pi_1(g(x))r, \pi_2(f(x)) + \pi_2(g(x))r) \\ &= (\pi_1(f(x)), \pi_2(f(x))) + (\pi_1(g(x)), \pi_2(g(x)))r \\ &= \varphi(f(x)) + \varphi(g(x))r\end{aligned}$$

Similarly, ψ is a homomorphism. Indeed, let $(g_1, h_1), (g_2, h_2) \in \text{Hom}_R(M, N_1) \oplus \text{Hom}_R(M, N_2)$, $x \in M$ and $r \in R$:

$$\begin{aligned}\psi((g_1, h_1) + (g_2, h_2)r)(x) &= \psi((g_1 + g_2r, h_1 + h_2r))(x) \\ &= i_1((g_1 + g_2r)(x)) + i_2((h_1 + h_2r)(x)) \\ &= i_1(g_1(x)) + i_1(g_2(x))r + i_2(h_1(x)) + i_2(h_2(x))r \\ &= \psi(g_1, h_1)(x) + \psi(g_2, h_2)(x)r\end{aligned}$$

Now, it's enough to show that φ is the inverse of ψ . Notice that for $(g, h) \in \text{Hom}_R(M, N_1) \oplus \text{Hom}_R(M, N_2)$, and $x \in M$:

$$\begin{aligned}\varphi \circ \psi(g, h)(x) &= \varphi(i_1(g(x)) + i_2(h(x))) \\ &= (\pi_1(i_1(g(x)) + i_2(h(x))), \pi_2(i_1(g(x)) + i_2(h(x)))) \\ &= ((\pi_1 \circ i_1) \circ g + (\pi_1 \circ i_2) \circ h, (\pi_2 \circ i_1) \circ g + (\pi_2 \circ i_2) \circ h)(x) \\ &= (g, h)(x)\end{aligned}$$

so $\varphi \circ \psi(g, h)$ is identity. Similarly, for $f \in \text{Hom}_R(M, N_1 \oplus N_2)$ and $x \in M$:

$$\begin{aligned}\psi \circ \varphi(f)(x) &= \psi(\pi_1 \circ f, \pi_2 \circ f)(x) \\ &= i_1 \circ \pi_1 \circ f(x) + i_2 \circ \pi_2 \circ f(x) \\ &= (i_1 \circ \pi_1 + i_2 \circ \pi_2) \circ f(x) = f(x)\end{aligned}$$

Hence, we have

$$\text{Hom}_R(M, N_1 \oplus N_2) \cong \text{Hom}_R(M, N_1) \oplus \text{Hom}_R(M, N_2)$$

(v) Consider $M = \mathbb{Z}/2\mathbb{Z}$, let $f \in \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z})$. Clearly, $f(0) = 0$. Notice

$$0 = f(0) = f(2) = f(1) + f(1) = 2f(1)$$

But $\mathbb{Z}$ is an integral domain so that $f(1) = 0$. Hence, $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) = 0$.

2 Problem sheet 2

Problem 2.1.

- (a) We have seen in the lecture that $\mathcal{E}$ splits on the right implies $\mathcal{E}$ splits. It's enough to show the other direction.
 $(\implies)$: Suppose $\mathcal{E}$ splits i.e. there is a commutative diagram of R -homomorphisms

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C & \longrightarrow & 0 \\ & & \downarrow \text{id}_A & & \downarrow \varphi & & \downarrow \text{id}_C & & \\ 0 & \longrightarrow & A & \xrightarrow{j} & A \oplus C & \xrightarrow{\pi} & C & \longrightarrow & 0 \end{array}$$

where j is the natural inclusion and π is the natural projection. Define

$$\begin{aligned} s : C &\rightarrow B \\ c &\mapsto \varphi^{-1} \begin{pmatrix} 0 \\ c \end{pmatrix} \end{aligned}$$

This is clearly a well-defined R -homomorphism. Notice that for any $c \in C$:

$$\pi \circ \varphi \circ s(c) = \pi \circ \varphi \circ \varphi^{-1} \begin{pmatrix} 0 \\ c \end{pmatrix} = \pi \begin{pmatrix} 0 \\ c \end{pmatrix} = c$$

i.e. $\pi \circ \varphi \circ s = \text{id}_C$. But since the diagram commutes, it follows that $\pi \circ \varphi = \text{id}_C \circ p = p$. Therefore,

$$\pi \circ \varphi \circ s = p \circ s = \text{id}_C$$

which means $\mathcal{E}$ splits on the right.

- (b) $(\implies)$: Suppose $\mathcal{E}$ splits, by part(a), we also know that $\mathcal{E}$ splits on the right. Thus, there exists some $s : C \rightarrow B$ such that $p \circ s = \text{id}_C$. Consider

$$\begin{aligned} r : B &\rightarrow A \\ b &\mapsto i^{-1}(b - s \circ p(b)) \end{aligned}$$

Then

$$\begin{aligned} r \circ i(a) &= i^{-1}(i(a) - s \circ p(i(a))) = a - s \circ p \circ i(a) \\ &= a - s \circ (p \circ i)(a) = a \end{aligned}$$

since exactness implies $p \circ i = 0$. Thus, $r \circ i = \text{id}_A$ i.e. $\mathcal{E}$ splits on the left.

$(\impliedby)$: Suppose $\mathcal{E}$ splits on the left i.e. there is an R -homomorphism $r : B \rightarrow A$ such that $r \circ i = \text{id}_A$. It's enough to check that $\varphi(b) := (r(b), p(b))$ is an isomorphism. Consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C & \longrightarrow & 0 \\ & & \downarrow \text{id}_A & & \downarrow \varphi & & \downarrow \text{id}_C & & \\ 0 & \longrightarrow & A & \xrightarrow{j} & A \oplus C & \xrightarrow{\pi} & C & \longrightarrow & 0 \end{array}$$

Notice that

$$\varphi \circ i(a) = (r \circ i(a), p \circ i(a)) = (a, 0) = j \circ \text{id}_A(a)$$

so

$$\begin{array}{ccc} A & \xrightarrow{i} & B \\ \downarrow \text{id}_A & & \downarrow \varphi \\ A & \xrightarrow{j} & A \oplus C \end{array}$$

commutes. Similarly,

$$\pi \circ \varphi(b) = \pi \circ (r(b), p(b)) = \text{id}_C \circ p(b)$$

so

$$\begin{array}{ccc} B & \xrightarrow{p} & C \\ \downarrow \varphi & & \downarrow \text{id}_C \\ A \oplus C & \xrightarrow{\pi} & C \end{array}$$

commutes. Hence, by the Five Lemma, we see that φ is an isomorphism.

Problem 2.2.

- (a) For $X = \Lambda$, we know that $\text{Hom}_\Lambda(X, A) \cong A$ then we know that there are isomorphisms $\varphi_1, \varphi_2, \varphi_3$ such that the following diagram

$$\begin{array}{ccccccc} \mathcal{E}_\Lambda = (0 & \longrightarrow & \text{Hom}_\Lambda(\Lambda, A) & \xrightarrow{i_*} & \text{Hom}_\Lambda(\Lambda, B) & \xrightarrow{p_*} & \text{Hom}_\Lambda(\Lambda, C) \longrightarrow 0) \\ & & \downarrow \varphi_1 & & \downarrow \varphi_2 & & \downarrow \varphi_3 \\ 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C \longrightarrow 0 \end{array}$$

commutes. Since

$$\mathcal{E} = (0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0)$$

is an exact sequence. It follows that $\mathcal{E}_\Lambda$ is an exact sequence as well.

- (b) For $X = \Lambda^n$, we know that

$$\text{Hom}_\Lambda(\Lambda^n, A) \cong \underbrace{A \oplus \cdots \oplus A}_n$$

Thus, there are isomorphisms $\varphi_1, \varphi_2, \varphi_3$ such that the following diagram

$$\begin{array}{ccccccc} \mathcal{E}_{\Lambda^n} = (0 & \longrightarrow & \text{Hom}_\Lambda(\Lambda^n, A) & \xrightarrow{i_*} & \text{Hom}_\Lambda(\Lambda^n, B) & \xrightarrow{p_*} & \text{Hom}_\Lambda(\Lambda^n, C) \longrightarrow 0) \\ & & \downarrow \varphi_1 & & \downarrow \varphi_2 & & \downarrow \varphi_3 \\ 0 & \longrightarrow & A^{\oplus n} & \xrightarrow{i^{\oplus n}} & B^{\oplus n} & \xrightarrow{p^{\oplus n}} & C^{\oplus n} \longrightarrow 0 \end{array}$$

commutes. Clearly, the sequence below is exact so $\mathcal{E}_{\Lambda^n}$ is exact as well.

(c) We first show that i_* is injective. But this is obvious since i is injective so

$$i_*(f) = 0 \implies i(f(x)) = 0 \text{ for all } x \in X$$

i.e. $f(x) = 0$ for all $x \in X$ so $f = 0$. $\ker i_* = \text{Im}(p_*)$ is an immediate consequence of the fact that $\mathcal{E}$ is exact. Now, it's enough to check that p_* is surjective. Since X is projective and p surjective. For every surjective homomorphism $p : B \rightarrow C$ and every homomorphism $g : X \rightarrow C$ there exists a lifting $h : X \rightarrow B$ such that $p \circ h = g$ i.e. the following diagram commutes

$$\begin{array}{ccc} & & B \\ & \nearrow h & \downarrow p \\ X & \xrightarrow{g} & C \end{array}$$

Then for any $g \in \text{Hom}_\Lambda(X, C)$, there exists some $h \in \text{Hom}_\Lambda(X, B)$ such that $p_*(h) = g$ i.e. p_* is surjective. Hence, we could conclude that $\mathcal{E}_X$ is exact.

Problem 2.3. It is enough to look at for any $n \in \mathbb{Z}_{\geq 0}$

$$0 \longrightarrow \text{Hom}_\Lambda(P_n, A) \xrightarrow{i_*} \text{Hom}_\Lambda(P_n, B) \xrightarrow{p_*} \text{Hom}_\Lambda(P_n, C) \longrightarrow 0$$

But it follows immediately from Q2(c) that the sequence above is exact.

3 Problem sheet 3

Problem 3.1.

- (i) Given that P and Q are projective, then there exists Λ -module K, M such that $K \oplus P$ and $M \oplus Q$ are free. Then

$$(K \oplus P) \oplus (M \oplus Q) \cong (K \oplus M) \oplus (P \oplus Q)$$

is free. Thus, $P \oplus Q$ is projective.

- (ii) Suppose P and $P \oplus Q$ are projective. Then there exists K, M such that $K \oplus P$ and $M \oplus (P \oplus Q)$ are free. But notice

$$(K \oplus P) \oplus (M \oplus (P \oplus Q)) \cong Q \oplus (K \oplus M \oplus P \oplus P)$$

is free. Thus, Q is projective.

Problem 3.2.

- (i) Reflexivity is clear, we can choose $S = T = \{0\}$. To see it's symmetric, suppose $K \sim L$ then there exists S, T projective such that $K \oplus S \cong L \oplus T$. If we choose, $S' = T$ and $T' = S$ then we have $L \oplus S' \cong K \oplus T'$ i.e. $L \sim K$. Now, to show transitivity, suppose $K \sim L$ and $L \sim M$ then there exists S, T, S', T' projective such that

$$K \oplus S \cong L \oplus T, \quad L \oplus S' \cong M \oplus T'$$

Then notice

$$(K \oplus S) \oplus S' \cong (L \oplus T) \oplus S' \cong (L \oplus S') \oplus T \cong (M \oplus T') \oplus T$$

i.e.

$$K \oplus (S \oplus S') \cong M \oplus (T' \oplus T).$$

We know that both $S \oplus S'$ and $T' \oplus T$ are projective by Q1. Therefore, $K \sim M$ so " $\sim$ " is an equivalence relation.

- (ii) Consider the exact sequence

$$0 \longrightarrow K \longrightarrow S \longrightarrow M \longrightarrow 0$$

and an obvious exact sequence

$$0 \longrightarrow 0 \longrightarrow T_0 \xrightarrow{\text{id}} T_0 \longrightarrow 0$$

We know that the direct sum of two exact sequences is also exact i.e.

$$0 \longrightarrow K \longrightarrow S \oplus T_0 \longrightarrow M \oplus T_0 \longrightarrow 0$$

Since we also know that

$$0 \longrightarrow L \longrightarrow T_1 \longrightarrow M \oplus T_0 \longrightarrow 0$$

By Schanuel's Lemma, we have $K \oplus T_1 \cong L \oplus (T_0 \oplus S)$. Since T_1 and $T_0 \oplus S$ are both projective, it follows that $K \sim L$.

(iii) It is enough to prove a generalized Schanuel's Lemma namely

$$K \oplus Q_n \oplus P_{n-1} \oplus \cdots \cong L \oplus P_n \oplus Q_{n-1} \oplus \cdots$$

Let $K_0 = M$, $L_0 = M$ and for $1 \leq i \leq n$, $K_i = \ker(P_i \rightarrow P_{i-1})$ and $L_i = \ker(Q_i \rightarrow Q_{i-1})$. And $K_{n+1} = K$, $L_{n+1} = L$. For $n = 0$ is really the usual Schanuel's Lemma. Assume that for $1 \leq i \leq n-1$, we have

$$K_i \oplus Q_i \oplus P_{i-1} \oplus \cdots \cong L_i \oplus P_i \oplus Q_{i-1} \oplus \cdots$$

Consider the syzygies

$$0 \longrightarrow K_{i+1} \longrightarrow P_{i+1} \longrightarrow K_i \longrightarrow 0$$

$$0 \longrightarrow L_{i+1} \longrightarrow Q_{i+1} \longrightarrow L_i \longrightarrow 0$$

Using the same idea from part(ii), (the direct sum of exact sequence is exact):

$$0 \longrightarrow K_{i+1} \longrightarrow P_{i+1} \oplus (Q_i \oplus P_{i-1} \oplus \cdots) \longrightarrow K_i \oplus (Q_i \oplus P_{i-1} \oplus \cdots) \longrightarrow 0$$

$$0 \longrightarrow L_{i+1} \longrightarrow Q_{i+1} \oplus (P_i \oplus Q_{i-1} \oplus \cdots) \longrightarrow L_i \oplus (P_i \oplus Q_{i-1} \oplus \cdots) \longrightarrow 0$$

i.e. by inductive hypothesis

$$0 \longrightarrow K_{i+1} \longrightarrow P_{i+1} \oplus (Q_i \oplus P_{i-1} \oplus \cdots) \longrightarrow K_i \oplus (Q_i \oplus P_{i-1} \oplus \cdots) \longrightarrow 0$$

$$0 \longrightarrow L_{i+1} \longrightarrow Q_{i+1} \oplus (P_i \oplus Q_{i-1} \oplus \cdots) \longrightarrow L_i \oplus (P_i \oplus Q_{i-1} \oplus \cdots) \longrightarrow 0$$

Hence, applying the usual Schanuel's Lemma we have

$$K_{i+1} \oplus Q_{i+1} \oplus (P_i \oplus Q_{i-1} \oplus \cdots) \cong L_{i+1} \oplus P_{i+1} \oplus (Q_i \oplus P_{i-1} \oplus \cdots)$$

i.e. $K \sim L$ since each Q_i, P_j are projective.

Problem 3.3. *Proof.* ($\implies$) : This part is easy, and have been done in the lecture.

($\impliedby$) : Let

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} P \longrightarrow 0$$

be any exact sequence. We want to show that this splits. This induces a long exact sequence:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}(P, N) & \xrightarrow{p_*} & \text{Hom}(B, N) & \xrightarrow{i_*} & \text{Hom}(A, N) \longrightarrow \\ & & & & \delta & & \\ & & & & \swarrow & & \\ & & H^1(P, N) & \xrightarrow{p_*} & H^1(B, N) & \longrightarrow & \cdots \end{array}$$

If $H^1(P, N) = 0$ for all N we have

$$0 \longrightarrow \text{Hom}(P, N) \xrightarrow{p_*} \text{Hom}(B, N) \xrightarrow{i_*} \text{Hom}(A, N) \longrightarrow 0$$

Let $N = A$ then

$$0 \longrightarrow \text{Hom}(P, A) \xrightarrow{p_*} \text{Hom}(B, A) \xrightarrow{i_*} \text{Hom}(A, A) \longrightarrow 0$$

is exact. So that i_* is surjective i.e. there exists some $s \in \text{Hom}(B, A)$ such that $i_*(s) = \text{id}_A$ so $s \circ i = \text{id}_A$ i.e.

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} P \longrightarrow 0$$

splits on the left. Thus, P is projective. $\square$

Problem 3.4.

(i) Let

$$0 \longrightarrow K \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

be an exact sequence. By the Mayer-Vietoris theorem, this induces a long exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}(M, N) & \longrightarrow & \text{Hom}(P_0, N) & \longrightarrow & \text{Hom}(K, N) \longrightarrow \\ & & & & \delta & & \\ & & \hookrightarrow & H^1(M, N) & \longrightarrow & H^1(P_0, N) & \longrightarrow \dots \end{array}$$

where for any $n \in \mathbb{N}$ we have

$$H^n(P_0, N) \longrightarrow H^n(K, N) \xrightarrow{\delta} H^{n+1}(M, N) \longrightarrow H^{n+1}(P_0, N)$$

Since P_0 is projective, we know that $H^1(P, N) = 0$ for all N i.e. $H^n(P, N) = 0$ for all $n \in \mathbb{N}$ and N . This means for $n \in \mathbb{N}$ we have

$$0 \longrightarrow H^n(K, N) \xrightarrow{\delta} H^{n+1}(M, N) \longrightarrow 0$$

be an exact sequence. By the property of short exact sequence, we know that δ is an isomorphism i.e. for each $n \in \mathbb{N}$

$$H^n(K, N) \cong H^{n+1}(M, N)$$

for all Λ -modules N .

- (ii) For this question, I believe we need to assume that $P_{n-1}, \dots, P_0$ are all projective. Let $K_0 = M$ and $K_n = K$, define the syzygies as K_i such that

$$0 \longrightarrow K_1 \longrightarrow P_0 \longrightarrow K_0 \longrightarrow 0$$

$$0 \longrightarrow K_2 \longrightarrow P_1 \longrightarrow K_1 \longrightarrow 0$$

$$\vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots$$

$$0 \longrightarrow K_n \longrightarrow P_{n-1} \longrightarrow K_{n-1} \longrightarrow 0$$

are exact sequences. Then by part(i), we know that

$$H^n(K_i, N) \cong H^{n+1}(K_{i-1}, N)$$

for all Λ -modules N . Thus,

$$\begin{aligned} H^{n+1}(M, N) &= H^{n+1}(K_0, N) \cong H^n(K_1, N) \cong H^{n-1}(K_2, N) \\ &\cong H^{n-2}(K_3, N) \cong \dots \cong H^1(K_n, N) = H^1(K, N) \end{aligned}$$

Since K is projective if and only if $H^1(K, N) = 0$ for all N , we are done.

4 Problem sheet 4

Problem 4.1.

(i) Consider

$$\begin{array}{ccc} A & \xrightarrow{i} & C \\ \downarrow f & & \downarrow \nu \\ B & \xrightarrow{j} & \underline{\lim}(f, i) \end{array}$$

where $\underline{\lim}(f, i) = B \oplus C / \text{Im}(-f \times i)$. Let i be injective, and let $b \in B$. Then $j(b) = 0$ iff

$$j(b) \in \text{Im}(-f \times i)$$

i.e. there exists some $a \in A$ such that

$$\begin{bmatrix} b \\ 0 \end{bmatrix} = \begin{bmatrix} -f(a) \\ i(a) \end{bmatrix}$$

i.e. $i(a) = 0$ and $b = -f(a)$. Since $\ker i$ is trivial, $a = 0$ so it follows that $b = -f(0) = 0$. Therefore, $j(b) = 0$ implies $b = 0$ i.e. j is injective.

(ii) Let f be surjective. If $y \in \underline{\lim}(f, i)$ then

$$y = \begin{bmatrix} b \\ c \end{bmatrix}$$

for some $b \in B, c \in C$ is an equivalence class. Since f is surjective, for any $b \in B$, there exists some $a \in A$ such that $f(a) = b$. Thus,

$$y = \begin{bmatrix} b \\ c \end{bmatrix} = \begin{bmatrix} f(a) \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ c - i(a) \end{bmatrix} = \nu(c - i(a)).$$

Since $c \in C$ and $i(a) \in C$, it follows that $c - i(a) \in C$ so ν is surjective.

Problem 4.2.

(i) By definition,

$$\underline{\lim}(\epsilon, i) = (\mathbb{Z}/2^m \oplus \mathbb{Z}/2^{n+k}) / \text{Im}(-\epsilon \times i)$$

Notice

$$|\mathbb{Z}/2^m \oplus \mathbb{Z}/2^{n+k}| = |\mathbb{Z}/2^m| |\mathbb{Z}/2^{n+k}| = 2^m 2^{n+k} = 2^{m+n+k}.$$

Consider the map

$$\begin{aligned} \varphi : \mathbb{Z}/2^n &\rightarrow \mathbb{Z}/2^m \oplus \mathbb{Z}/2^{n+k} \\ a &\mapsto \begin{bmatrix} -f(a) \\ i(a) \end{bmatrix} \end{aligned}$$

then by the First Isomorphism Theorem, $|\text{Im}(\varphi)| = |\mathbb{Z}/2^n|/|\ker \varphi| = 2^n/|\ker \varphi|$. Thus, it's enough to show that $\ker \varphi$ is injective i.e. both i and ϵ are injective. But this should be clear for example

$$2^k x \equiv 0 \pmod{2^{n+k}} \implies 2^n \mid x \text{ i.e. } [x]_n = 0$$

$$\text{Thus, } |\varprojlim(\epsilon, i)| = 2^{m+n+k}/2^n = 2^{m+k}.$$

Problem 4.3. We denote the elements of $\mathbb{Z}/8$ by

$$\{0, 1, 2, 3, 4, 5, 6, 7\}$$

- (i) i is injective since $\ker(\times 4 \text{ on } \mathbb{Z}/8) = \{0\}$ on $\mathbb{Z}/2$. ϵ is clearly surjective. $\text{Im}(i) = \{0, 4\}$ while $\ker(\epsilon) = \{0, 4\}$. Thus, the sequence is exact.
- (ii) Almost the same as above.
- (iii) $i \circ \eta$ really means $i \circ \eta(x) = 4$ if x is "odd" and $i \circ \eta(x) = 0$ if x is "even" i.e. $i \circ \eta$ is multiplication by 4 on $\mathbb{Z}/8$. Similarly, $j \circ \epsilon$ is multiplication by 2 on $\mathbb{Z}/8$. Thus, $\ker(j \circ \epsilon) = \{0, 4\} = \text{Im}(i \circ \eta)$. And $\text{Im}(j \circ \epsilon) = \{0, 2, 4, 6\} = \ker(i \circ \eta)$. Lastly, $\text{Im}(i \circ \eta) = \{0, 4\} = \ker(\epsilon)$. Thus, we know that the sequence

$$0 \longrightarrow \mathbb{Z}/8 \xrightarrow{i \circ \eta} \mathbb{Z}/8 \xrightarrow{j \circ \epsilon} \mathbb{Z}/8 \xrightarrow{i \circ \eta} \mathbb{Z}/8 \xrightarrow{\epsilon} \mathbb{Z}/4 \longrightarrow 0$$

is exact.

Problem 4.4.

- (a) We apply $\text{Hom}(\cdot, \mathbb{Z}/8)$ to part(iii) above to get

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}(\mathbb{Z}/4, \mathbb{Z}/8) & \xrightarrow{\epsilon^*} & \text{Hom}(\mathbb{Z}/8, \mathbb{Z}/8) & \xrightarrow{(i \circ \eta)^*} & \text{Hom}(\mathbb{Z}/8, \mathbb{Z}/8) \longrightarrow \\ & & & & (j \circ \epsilon)^* & & \\ & & & & & \swarrow & \\ & & & & & \text{Hom}(\mathbb{Z}/8, \mathbb{Z}/8) & \xrightarrow{(i \circ \eta)^*} \text{Hom}(\mathbb{Z}/8, \mathbb{Z}/8) \longrightarrow 0 \end{array}$$

This is equivalent to

$$0 \longrightarrow \mathbb{Z}/8 \xrightarrow{i \circ \eta} \mathbb{Z}/8 \xrightarrow{j \circ \epsilon} \mathbb{Z}/8 \xrightarrow{i \circ \eta} \mathbb{Z}/8 \longrightarrow \dots$$

Then $H^0(\mathbb{Z}/4, \mathbb{Z}/8) = \ker(i \circ \eta) = \{0, 2, 4, 6\} = \mathbb{Z}/4$. But for $k \geq 1$, since the sequence is exact, it follows that $H^k(\mathbb{Z}/4, \mathbb{Z}/8) = 0$.

- (b) Applying $\text{Hom}(\cdot, \mathbb{Z}/2)$ we have

$$0 \longrightarrow \mathbb{Z}/2 \xrightarrow{i \circ \eta} \mathbb{Z}/2 \xrightarrow{j \circ \epsilon} \mathbb{Z}/2 \xrightarrow{i \circ \eta} \mathbb{Z}/2 \longrightarrow \dots$$

Both $i \circ \eta$ and $j \circ \epsilon$ are all zero map. Thus, $H^0(\mathbb{Z}/4, \mathbb{Z}/2) = \mathbb{Z}/2$. For all $k \geq 1$, $H^k(\mathbb{Z}/4, \mathbb{Z}/2) = \ker(0)/0 = \mathbb{Z}/2$.

(c) Applying $\text{Hom}(\cdot, \mathbb{Z}/4)$ we have

$$0 \longrightarrow \mathbb{Z}/4 \xrightarrow{i \circ \eta} \mathbb{Z}/4 \xrightarrow{j \circ \epsilon} \mathbb{Z}/4 \xrightarrow{i \circ \eta} \mathbb{Z}/4 \longrightarrow \dots$$

Notice $i \circ \eta$ is the zero map while $j \circ \epsilon$ is still multiplication by 2 on $\mathbb{Z}/4$. For $k = 0$, we have $H^0(\mathbb{Z}/4, \mathbb{Z}/4) = \ker(i \circ \eta) \cong \mathbb{Z}/4$. For k odd, $H^k(\mathbb{Z}/4, \mathbb{Z}/4) \cong \ker(j \circ \epsilon)/\text{Im}(i \circ \eta) = \{0, 2\}/0 \cong \mathbb{Z}/2$. And for k even, $H^k(\mathbb{Z}/4, \mathbb{Z}/4) \cong \ker(i \circ \eta)/\text{Im}(j \circ \epsilon) = \mathbb{Z}/4/\{0, 2\} \cong \mathbb{Z}/2$.

5 Problem sheet 5

Problem 5.1.

- (i) For $r \in \mathbb{N}$, define

$$B_r := \begin{cases} \ker(\eta^r), & \text{if } r \geq 1 \\ 0 & \text{if } r = 0 \end{cases}$$

Then for $x \in B_r$, we have $\eta^r(x) = 0$ so that $\eta^{r+1}(x) = \eta(\eta^r(x)) = \eta(0) = 0$ i.e. $x \in B_{r+1}$ which means $B_r \subset B_{r+1}$. Since η is surjective, η^r is surjective as well. Let $K = \ker(\eta)$, then the map $\eta^k : B_{r+1} \rightarrow \ker(\eta)$ is surjective. By the First Isomorphism Theorem, we have

$$B_{r+1} / \ker(\eta^k) \cong \ker(\eta)$$

i.e. $B_{r+1}/B_r \cong K$.

- (ii) Suppose K is non-trivial, then for each r we have $B_r \subsetneq B_{r+1}$. Hence, we have generated an ascending sequence of submodules

$$B_0 \subsetneq B_1 \subsetneq B_2 \subsetneq \dots$$

which does not terminate. This means A is non-Noetherian.

Problem 5.2. Let $f : A \rightarrow A \oplus S$ be an isomorphism and $\pi : A \oplus S \rightarrow A$ be the natural projection map. Define $\varphi = \pi \circ f : A \rightarrow A$ then

$$\ker \varphi = f^{-1}(\ker \pi) \cong \ker \pi \cong S$$

In particular, $\ker \varphi$ is a submodule of A . Consider the sequence of submodules

$$\ker \varphi \subseteq \ker \varphi^2 \subseteq \dots$$

Since A is Noetherian, there exists some $N \in \mathbb{N}$ such that $\ker \varphi^r = \ker \varphi^N$ for all $r \geq N$. Let $x \in \ker \varphi$, since both f and π are surjective, φ is surjective as well. In particular, φ^N is surjective, thus there exists some y such that $\varphi^N(y) = x$. Then $\varphi^{N+1}(y) = \varphi(\varphi^N(y)) = \varphi(x) = 0$. This implies $y \in \ker \varphi^{N+1} = \ker \varphi^N$ so $x = \varphi^N(y) = 0$. Hence, we see that $\ker \varphi = \{0\}$. In this case, $S = 0$.

Problem 5.3. Let $\Lambda^m \cong \Lambda^n$, suppose WLOG that $m < n$ then we have

$$\Lambda^m \cong \Lambda^n \cong \Lambda^m \oplus \Lambda^{n-m}$$

Since Λ is a right Noetherian ring, it follows that Λ^m is right Noetherian as well. Thus, by Q2 we have $\Lambda^{n-m} = 0$ i.e. $n = m$.

Problem 5.4. Let R be a non-trivial commutative ring. Suppose every R -module is free. If R is not a field then there exists some $a \in R$ not invertible, and consider $R/(a)$ which is free and also non-empty. Then there exists a basis $\{e_1, \dots, e_k\}$ for $R/(a)$. Since (a) is non-empty, there exists some $a' \in (a) \setminus \{0\}$ so that $e_j \cdot a' = 0$ for all j . But this contradicts to the fact that $\{e_1, \dots, e_k\}$ are linearly independent. If $R/(a) = \text{span}\{e_1\}$ then we have $e_1 \cdot a = 0$, but since $\{e_1\}$ is a basis, it follows that $a = 0$. And hence, R is a field.

6 Problem sheet 6

Problem 6.1.

- (i) Suppose $(a, c) = 1$ and $(b, c) = 1$ then there exists $\mu_1, \mu_2, \lambda_1, \lambda_2 \in R$ such that

$$a\mu_1 + c\lambda_1 = 1, \quad b\mu_2 + c\lambda_2 = 1$$

This implies

$$(a\mu_1 + c\lambda_1)(b\mu_2 + c\lambda_2) = 1$$

i.e.

$$ab\mu_1\mu_2 + ca\mu_1\lambda_2 + cb\lambda_1\mu_2 + c^2\lambda_1\lambda_2 = 1$$

In particular,

$$(ab)\underbrace{(\mu_1\mu_2)}_{\in R} + c\underbrace{(a\mu_1\lambda_2 + b\lambda_1\mu_2 + c\lambda_1\lambda_2)}_{\in R} = 1$$

That is, $(ab, c) = 1$.

- (ii) This follows immediately by part(i) and induction.

Problem 6.2. For each i , $(a_i, c) = 1$ implies $(a_i^{e_i}, c) = 1$ for all i by Q1(ii). Now, by Q1(i) we have

$$(a_1^{e_1} a_2^{e_2} \cdots a_n^{e_n}, c) = 1$$

Clearly, $(a, b) = (b, a)$ for any $a, b \in R$ so it follows that

$$(a_1^{e_1} a_2^{e_2} \cdots a_n^{e_n}, c^{e_{n+1}}) = 1$$

Problem 6.3. Recall from the lecture: if $a = bc$ and $(b, c) = 1$ then $\Theta_a(M) = \Theta_b(M) \dot{+} \Theta_c(M)$. Let $a = p_1^{e_1} \cdots p_n^{e_n}$ where $(p_i, p_j) = 1$ for $i \neq j$. For any j :

$$\underbrace{(p_1^{e_1} \cdots p_n^{e_n})}_{\text{without } p_j^{e_n}}, p_j^{e_j} = 1$$

We then know that

$$\Theta_a(M) = \Theta_{p_1^{e_1}}(M) \dot{+} \Theta_{a_1}(M)$$

where $a_1 = p_2^{e_2} \cdots p_n^{e_n}$. We can do this repeatedly to deduce

$$\Theta_a(M) = \Theta_{p_1^{e_1}}(M) \dot{+} \cdots \dot{+} \Theta_{p_n^{e_n}}(M).$$

Problem 6.4. Suppose $p_1, \dots, p_m$ are all distinct primes and $q_1, \dots, q_n$ are all distinct primes as well. Since $p_1, \dots, p_m, q_1, \dots, q_n$ are all primes in R . And $p_1 \cdots p_m = q_1 \cdots q_n$. In particular, " $p_1 \mid q_1 \cdots q_n$ " which implies there exists some $\lambda_1 \in R^*$ such that $p_1 \lambda_1 = q_{\sigma(1)}$. Similarly, $p_j \lambda_j = q_{\sigma(j)}$. Notice that $\sigma(i) \neq \sigma(j)$ for $i \neq j$. Otherwise, for example $p_1 \lambda_1 = p_2 \lambda_2$ then p_1, p_2 are not

distinct primes. We first show that $m = n$, if $n > m$ then σ is clearly a bijection and after reordering ($q_1 = p_1 \lambda_1$) we have

$$p_1 \cdots p_m = p_1 \cdots p_m q_{m+1} \cdots q_n \lambda_1 \cdots \lambda_m$$

Since R is an integral domain, we know that

$$1 = q_{m+1} \cdots q_n \lambda_1 \cdots \lambda_m$$

But this implies $q_{m+1} \cdots q_n \in R^*$, a contradiction. Similar argument shows $m = n$. And it then follows immediately that there exists a bijection $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ such that, for all i

$$q_i = p_{\sigma(i)} \lambda_i \text{ where } \lambda_i \in R^*.$$

7 Problem sheet 7

Problem 7.1. We have given that $\text{Hom}_\Lambda(M, \Lambda)$ is a left Λ -module under the action

$$\begin{aligned}\diamond : \Lambda \times \text{Hom}_\Lambda(M, \Lambda) &\rightarrow \text{Hom}_\Lambda(M, \Lambda) \\ (\lambda \diamond f)(x) &= \lambda \cdot f(x)\end{aligned}$$

We want to show that the evaluation mapping

$$\begin{aligned}\text{ev} : \text{Hom}_\Lambda(\Lambda_{\text{right}}, \Lambda) &\rightarrow \Lambda_{\text{left}} \\ \text{ev}(f) &= f(1)\end{aligned}$$

is an isomorphism of left Λ -modules. We first check that ev is a homomorphism, clearly if $f, g \in \text{Hom}_\Lambda(\Lambda_{\text{right}}, \Lambda)$ then

$$\text{ev}(f + g) = (f + g)(1) = f(1) + g(1) = \text{ev}(f) + \text{ev}(g)$$

And for any $\lambda \in \Lambda$ we have

$$\text{ev}(\lambda \diamond f) = \lambda \cdot f(1) = \lambda \cdot \text{ev}(f).$$

To check injectivity, if $\text{ev}(f) = 0$ i.e. $f(1) = 0$. Then for any $x \in \Lambda$ we have

$$f(x) = f(1 \cdot x) = f(1) \cdot x$$

Thus, if $f(1) = 0$ we have $f(x) = 0$ for all x i.e. f is the zero map. Therefore, we see that $\ker(\text{ev}) = \{0\}$.

To show surjectivity, let $\alpha \in \Lambda_{\text{left}}$. Then define $f_\alpha(x) = \alpha \cdot x$ which is clearly a right module map. Then $\text{ev}(f_\alpha) = f_\alpha(1) = \alpha \cdot 1 = \alpha$. Therefore, ev is surjective. Thus, we have shown that ev is indeed an isomorphism of left Λ -modules.

Problem 7.2.

- (i) Let $\lambda_1, \lambda_2 \in \Lambda$, then $\underline{a}(\lambda_1 + \lambda_2) = a \cdot (\lambda_1 + \lambda_2) = a \cdot \lambda_1 + a \cdot \lambda_2 = \underline{a}(\lambda_1) + \underline{a}(\lambda_2)$. And if $r \in \Lambda$ then $\underline{a}(r\lambda) = a \cdot r\lambda = r(a \cdot \lambda) = r\underline{a}(\lambda)$. Similar argument shows that $\hat{a}$ is also a homomorphism of Λ -module.

- (ii) We want to show the following diagram commutes

$$\begin{array}{ccc}\text{Hom}_\Lambda(\Lambda, \Lambda) & \xrightarrow{\hat{a}} & \text{Hom}_\Lambda(\Lambda, \Lambda) \\ \downarrow \text{ev} & & \downarrow \text{ev} \\ \Lambda & \xrightarrow{\underline{a}} & \Lambda\end{array}$$

Let $f \in \text{Hom}_\Lambda(\Lambda, \Lambda)$ then $\text{ev} \circ \hat{a}(f) = \text{ev}(a \cdot f) = a \cdot f(1)$. Similarly, $\underline{a} \circ \text{ev}(f) = \underline{a}(f(1)) = a \cdot f(1)$. Thus, it follows that $\text{ev} \circ \hat{a} = \underline{a} \circ \text{ev}$ i.e. the diagram commutes.

(iii) Let $(\underline{a})^*(f) = f \circ \underline{a}$. Let $x \in \Lambda$ then

$$f \circ \underline{a}(x) = f(a \cdot x) = a \cdot f(x) = \hat{a}(f)(x)$$

i.e. $\hat{a} = (\underline{a})^*$.

(iv) We are given that

$$\Lambda \xrightarrow{\underline{b}} \Lambda \xrightarrow{\underline{a}} \Lambda$$

is exact. Then we have the following commutative diagram:

$$\begin{array}{ccccc} \mathrm{Hom}_\Lambda(\Lambda, \Lambda) & \xrightarrow{\hat{b}} & \mathrm{Hom}_\Lambda(\Lambda, \Lambda) & \xrightarrow{\hat{a}} & \mathrm{Hom}_\Lambda(\Lambda, \Lambda) \\ \downarrow \mathrm{ev} & & \downarrow \mathrm{ev} & & \downarrow \mathrm{ev} \\ \Lambda & \xrightarrow{\underline{b}} & \Lambda & \xrightarrow{\underline{a}} & \Lambda \end{array}$$

Since $\hat{a} \circ \hat{b} = (\underline{a})^*(\underline{b})^* = (\underline{b})^*(\underline{a})^* = (\underline{a} \circ \underline{b})^* = 0$. Thus, clearly we have $\text{Im}(\hat{b}) \subseteq \ker(\hat{a})$. Now, suppose that $f \in \ker(\hat{a})$ then $\text{ev} \circ \hat{a}(f) = \underline{a} \circ \text{ev}(f)$. Therefore, $\text{ev}(f) \in \ker(\underline{a})$ i.e. $\text{ev}(f) \in \text{Im}(\underline{b})$. This means there exists some $x \in \Lambda$ such that $\text{ev}(f) = \underline{b}(x)$. Since ev is surjective, there exists some $g \in \text{Hom}_\Lambda(\Lambda, \Lambda)$ such that $\text{ev}(g) = x$. Thus, $\text{ev}(f) = \underline{b} \circ \text{ev}(g)$. But this means $\text{ev}(f) = \text{ev} \circ \hat{b}(g)$. Injectivity of ev implies $f = \hat{b}(g)$ i.e. $f \in \text{Im}(\hat{b})$. This implies $\text{Im}(\hat{b}) = \ker(\hat{a})$ i.e.

$$\mathrm{Hom}_{\Lambda}(\Lambda, \Lambda) \xrightarrow{\hat{b}} \mathrm{Hom}_{\Lambda}(\Lambda, \Lambda) \xrightarrow{\hat{a}} \mathrm{Hom}_{\Lambda}(\Lambda, \Lambda)$$

is exact.

Problem 7.3. We are given an exact sequence

$$\cdots \longrightarrow \Lambda \xrightarrow{\underline{a}} \Lambda \xrightarrow{\underline{b}} \Lambda \xrightarrow{\underline{a}} \Lambda \xrightarrow{\epsilon} M \longrightarrow 0$$

Then we apply $\text{Hom}_\Lambda(\cdot, \Lambda)$ we get a sequence of cohomology:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathrm{Hom}_\Lambda(\Lambda, \Lambda) & \xrightarrow{\hat{a}} & \mathrm{Hom}_\Lambda(\Lambda, \Lambda) & \xrightarrow{\hat{b}} & \mathrm{Hom}_\Lambda(\Lambda, \Lambda) \longrightarrow \\
& & & & \hat{a} & & \\
& & \mathrm{Hom}_\Lambda(\Lambda, \Lambda) & \xrightarrow{\hat{b}} & \mathrm{Hom}_\Lambda(\Lambda, \Lambda) & \xrightarrow{\hat{a}} & \dots
\end{array}$$

By Q2, we know that the sequence is exact. Therefore, $H^k(M, \Lambda) = 0$ for all $k \geq 1$.

Problem 7.4.

(i) Consider the following sequence

$$\cdots \longrightarrow \Lambda \xrightarrow{(z-1)} \Lambda \xrightarrow{\Sigma} \Lambda \xrightarrow{(z-1)} \Lambda \xrightarrow{\epsilon} \mathbb{Z} \longrightarrow 0$$

where

$$\epsilon \left(\sum_{r=0}^{n-1} a_r t^r \right) = \sum_{r=0}^{n-1} a_r \text{ and } \Sigma = \sum_{r=0}^{n-1} t^r$$

and $\Lambda = \mathbb{Z}[C_n]$. Obviously by definition, ϵ is surjective. And in addition if $f \in \text{Im}(\underline{z-1})$ we have

$$f = (t-1) \sum_{r=0}^{n-1} a_r t^r = \sum_{r=1}^{n-1} (a_{r-1} - a_r) t^r + (a_{n-1} - a_0)$$

Thus,

$$\epsilon(f) = \sum_{r=1}^{n-1} (a_{r-1} - a_r) + (a_{n-1} - a_0) = 0$$

i.e. $\text{Im}(\underline{z-1}) \subseteq \ker(\epsilon)$. On the other hand, if $f \in \ker(\epsilon)$ we have

$$\sum_{r=0}^{n-1} a_r = 0$$

i.e. $f(1) = 0$. This implies

$$(t-1) \mid \sum_{r=0}^{n-1} a_r t^r$$

so $\ker(\epsilon) \subseteq \text{Im}(\underline{z-1})$. Thus, $\ker(\epsilon) = \text{Im}(\underline{z-1})$. Now, let $f \in \text{Im}(\underline{\Sigma})$ then

$$(t-1) \left(\sum_{r=0}^{n-1} t^r \right) \left(\sum_{r=0}^{n-1} a_r t^r \right) = (t^n - 1) \left(\sum_{r=0}^{n-1} a_r t^r \right) = 0$$

Thus, $f \in \ker(\underline{z-1})$ i.e. $\text{Im}(\underline{\Sigma}) \subseteq \ker(\underline{z-1})$. Conversely, if $f \in \ker(\underline{z-1})$ then

$$(t-1) \sum_{r=0}^{n-1} a_r t^{r-1} = 0$$

$$\sum_{r=1}^{n-1} (a_{r-1} - a_r) t^r + (a_{n-1} - a_0) = 0$$

so $a_0 = a_1 = \dots = a_{n-1}$. This means

$$f = a_0 \sum_{r=0}^{n-1} t^{r-1} \in \text{Im}(\underline{\Sigma})$$

Therefore, $\text{Im}(\underline{\Sigma}) = \ker(\underline{z-1})$. It remains to show that $\ker(\epsilon) = \ker(\underline{\Sigma})$ so that $\text{Im}(\underline{z-1}) = \ker(\underline{\Sigma})$. Notice

$$\left(\sum_{r=0}^{n-1} t^r \right) \left(\sum_{r=0}^{n-1} a_r t^r \right) = \left(\sum_{i=0}^{n-1} a_i \right) \left(\sum_{r=0}^{n-1} t^r \right)$$

That is

$$\Sigma \cdot f = 0 \text{ if and only if } \epsilon(f) = 0$$

which means $\ker(\epsilon) = \ker(\underline{\Sigma})$. Hence, the sequence is exact.

(ii) It follows immediately by Q3 and part(i) that $H^k(\mathbb{Z}, \mathbb{Z}[C_n]) = 0$ for all $k \geq 1$.

(iii) This also follows immediately since the following sequence is exact:

$$\cdots \longrightarrow \Lambda \xrightarrow{(z-1)} \Lambda \xrightarrow{\underline{\Sigma}} \Lambda \xrightarrow{(z-1)} \ker(\epsilon) \longrightarrow 0$$

so $H^k(\ker(\epsilon), \mathbb{Z}[C_n]) = 0$ for all $k \geq 1$.

8 Revision Questions

Problem 8.1. A module P is called projective when there exists $\tilde{\varphi}$ such that the following diagram commutes:

$$\begin{array}{ccc} & P & \\ \tilde{\varphi} \swarrow & & \downarrow \varphi \\ X & \xrightarrow{\eta} & M \end{array}$$

where η is surjective. In this case, $\eta \circ \tilde{\varphi} = \varphi$.